A lightweight vision-language model for Earth Observation

Remote sensing text-to-image retrieval with interpretable experts

Challenge 12

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Why this project matters

Remote sensing challenge

- Earth observation data is rich, but often difficult to query without technical expertise.
- Users need both directions of retrieval: text → image and image → text.
- For practical deployment, the model should stay lightweight, explainable, and easy to inspect.

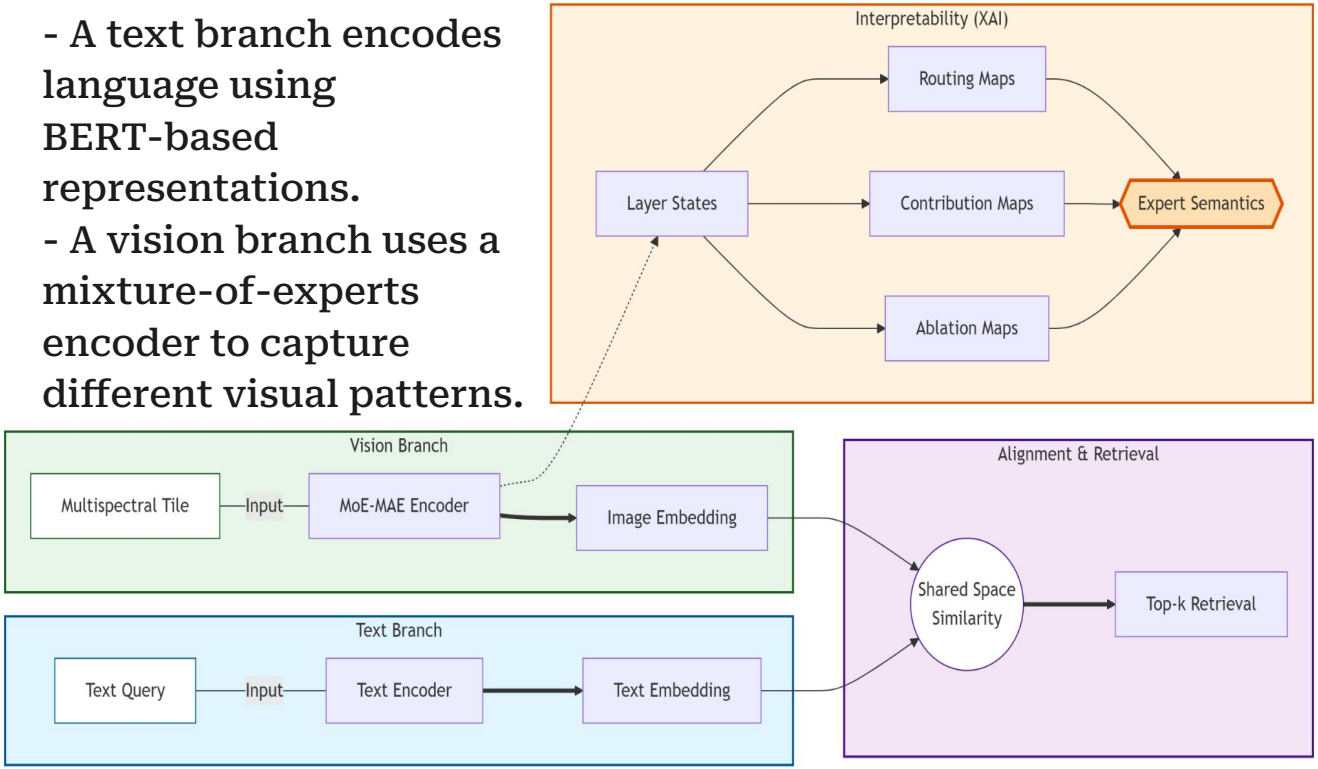
What are we doing here?

Objectives

1. Natural language retrieval
2. Search image by text vs. Search text by image
3. Integrating BERT (text only model) to use it for VL tasks
4. Interpretation of the experts in the vision encoder (Mixture-of-experts)

We are not only improving retrieval quality; we are making EO tools more usable for non-experts.

- A text branch encodes language using BERT-based representations.
- A vision branch uses a mixture-of-experts encoder to capture different visual patterns.



- Both branches are projected into a shared space and trained via CLIP style so text and images can be matched.
- Expert semantics help us inspect what different parts of the vision model are learning.

⭐Model architecture
⭐Loss function

Key achievements:

- The MoE-MAE is a light-weight architecture that balances efficiency and representational capacity
- The language representational capabilities of sBERT along with the contrastive alignment are exploited and the overall model performs well to unseen text queries

Retrieval and Interpretability Experiments

Retrieval Output Interpretation:

Analyse more than top-1 retrievals of open-vocabulary queries

Interpretability:

Where do tokens go? ->**Routing assessment map**

Idea: Explore the first layer that captures high-level patterns in the data to see whether the routing lead to experts that focus on images of specific classes (e.g. water, land etc.)

Demo Application

1. User input

Text prompt
or
example image

2. Model reasoning

Encode both modalities
Align them in a shared space
Use expert semantics for interpretation

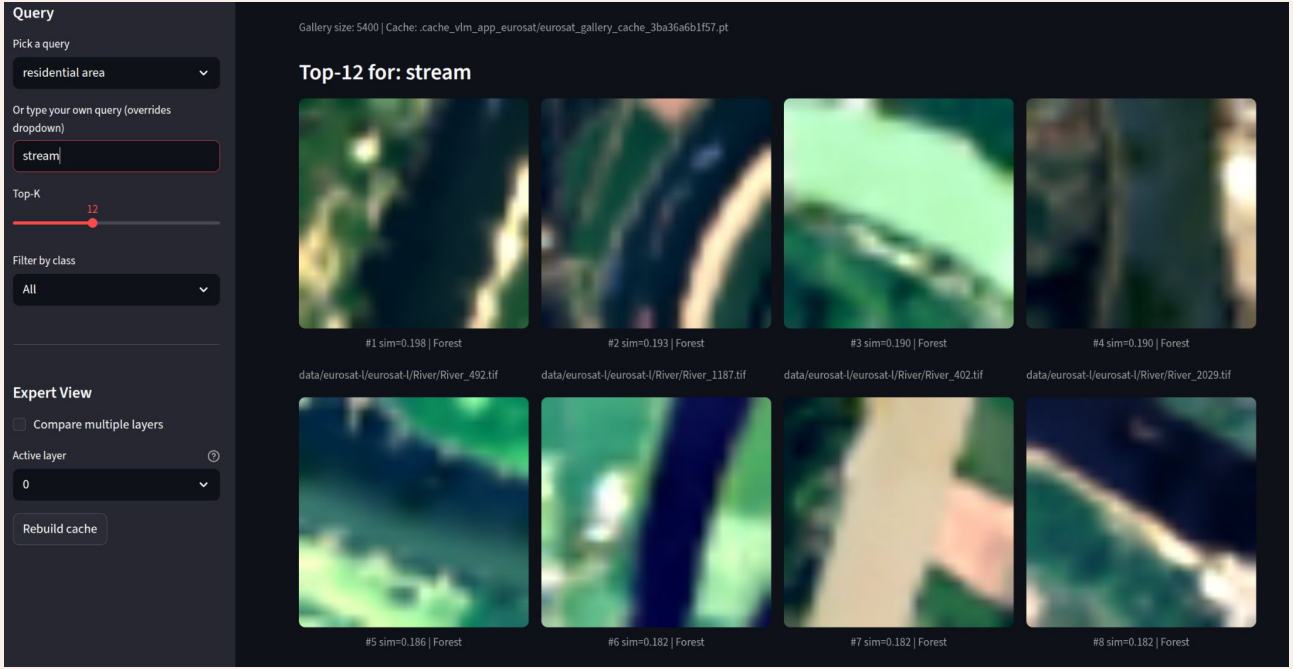
3. Ranked output

Most relevant images or captions returned to the user +
A map that indicates the top-1 expert for each patch

This is where lightweight + interpretable matters.

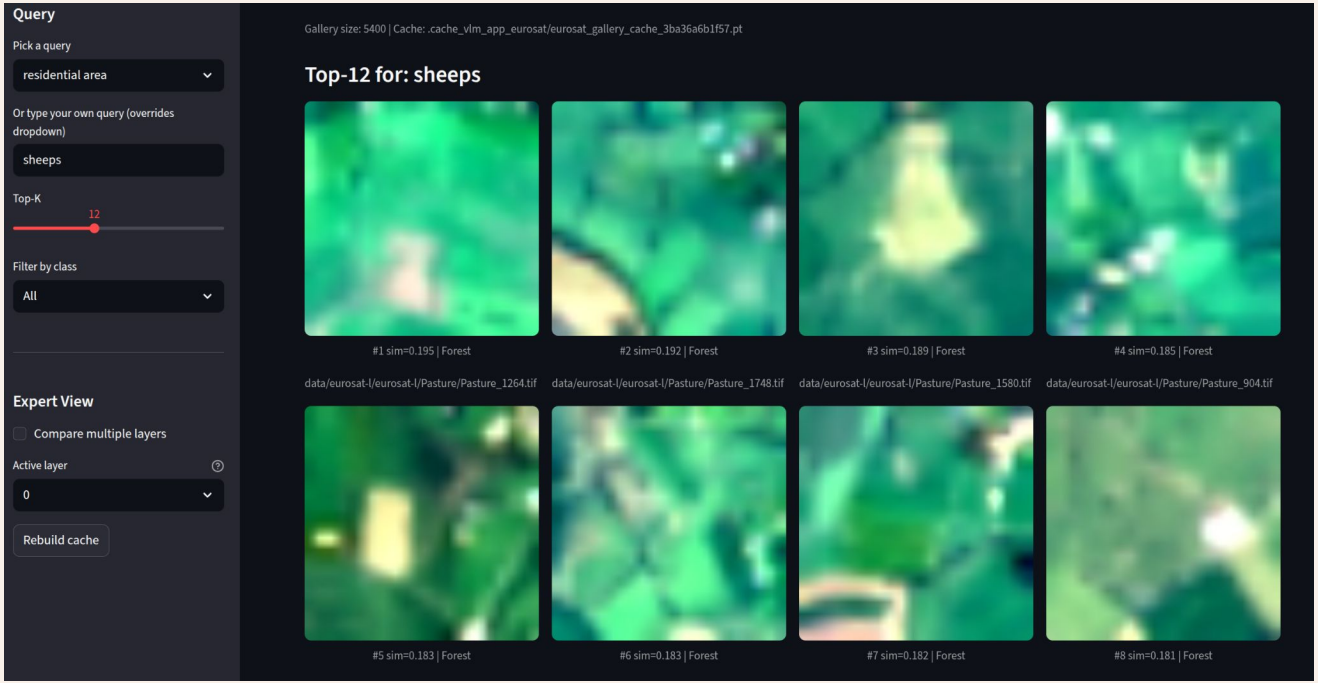
What can a user ask?

"Find satellite images showing a stream."



What can a user ask?

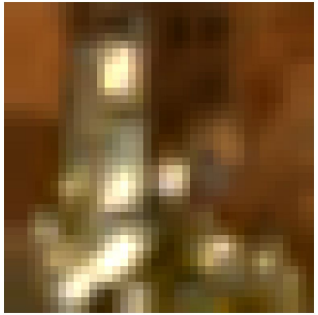
"Find satellite images showing sheeps."



What can a user ask?

"Given this image, retrieve the most relevant description or report."

1. Grazing Pasture Land
2. Orchards or Permanent Crops
3. Mosaic of Crop Fields

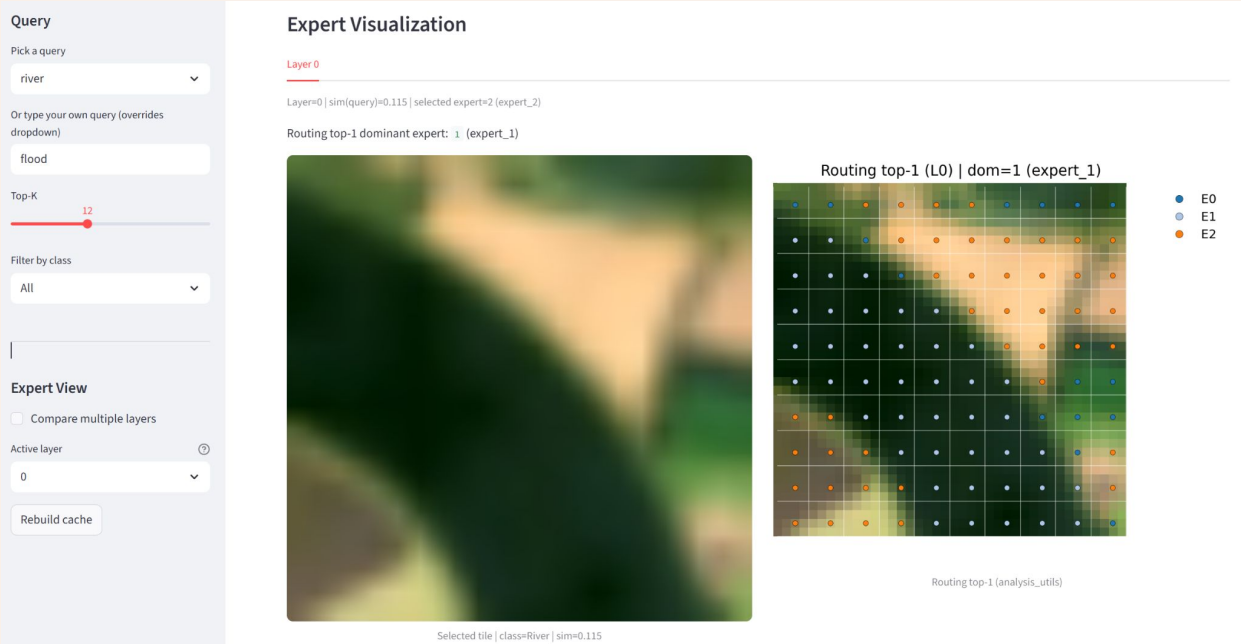


1. Industrial Buildings and Warehouses
2. Industrial Area Connected to Transport Infrastructure
3. Residential Area Near a Highway

1. River Crossing Vegetated Land
2. Major Road Corridor
3. Inland water body

What can a user ask?

"Which expert is focusing on vegetation, water, or urban patterns?"



Impact and DEI exercise

Impact

Why?

- Supports environmental monitoring, agriculture, and urban planning.
- Helps decision-makers access EO information faster.
- Turns retrieval into a practical entry point for downstream tasks.

Inclusion

Who is it for?

- Relevant for public authorities, scientists, and NGOs.
- Especially valuable for local decision-makers without EO expertise.
- Can lower the barrier for communities with limited access to EO tools or digital infrastructure.

Potential improvements?

- Add human-in-the-loop feedback to refine expert behavior.
- Increase the number of experts, scale up the model, add more modalities.

Thank you!

Supporting Partners: